

Credit Card Behaviour Score Prediction

Using Classification and Risk-Based
Techniques

Overview

- Goal: Predict credit card default using classification techniques.
- Dataset includes demographic, payment, and billing features.
- End-use: Help financial institutions assess credit risk effectively.

Modeling Strategy

Approach:

1. Data Preparation

- Cleaned missing values
- Balanced dataset using **SMOTETomek**

2. Feature Engineering

- Focused on **payment behaviour, credit utilization, overdue patterns**
- Created new features → **Longest Overdue Streak, Utilization Ratios**

3. Model Building

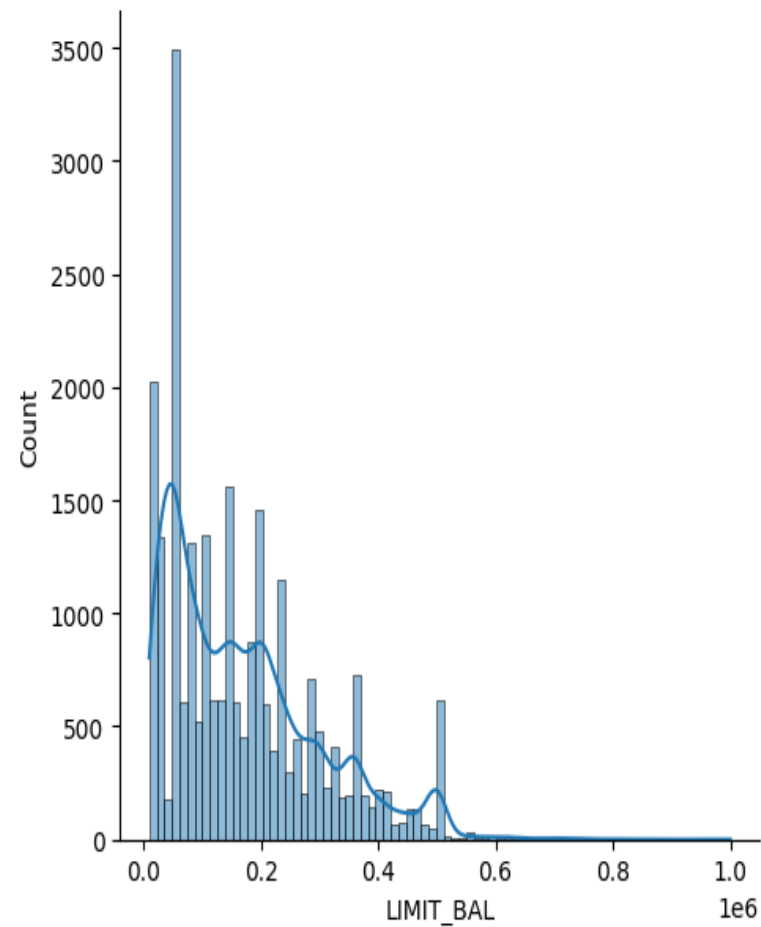
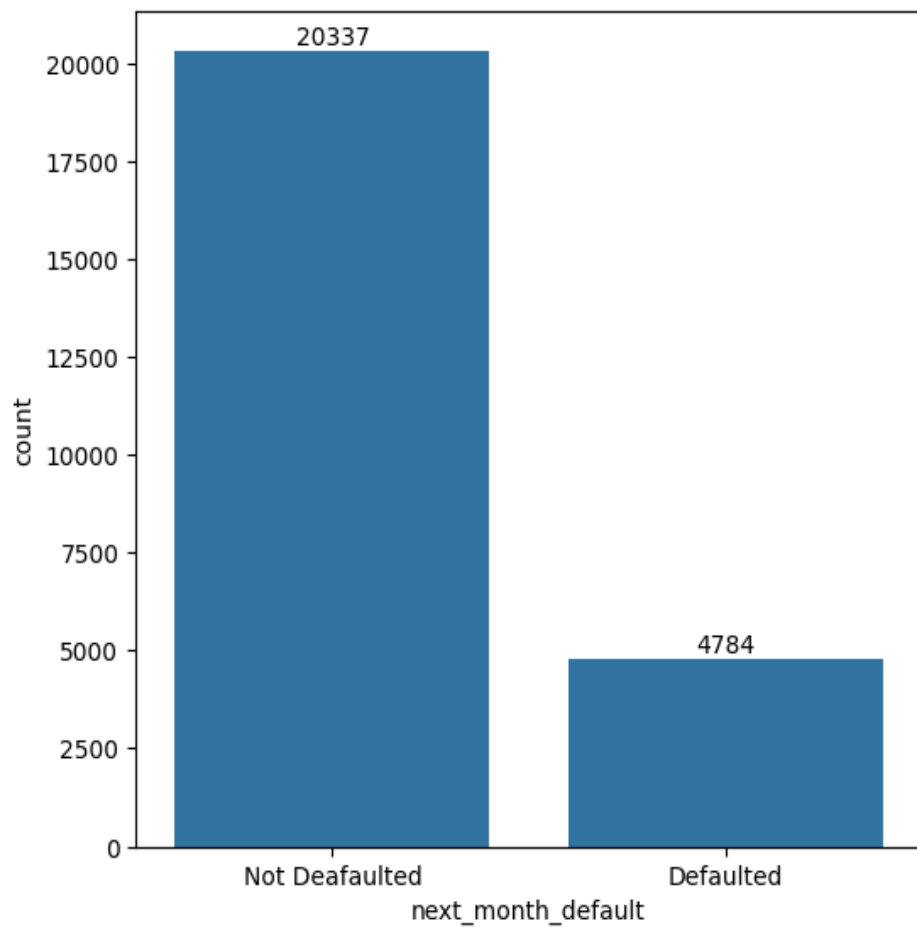
- Used **Random Forest, LightGBM, XGBoost**
- Tuned using **cross-validation & threshold optimization**

4. Evaluation Focus

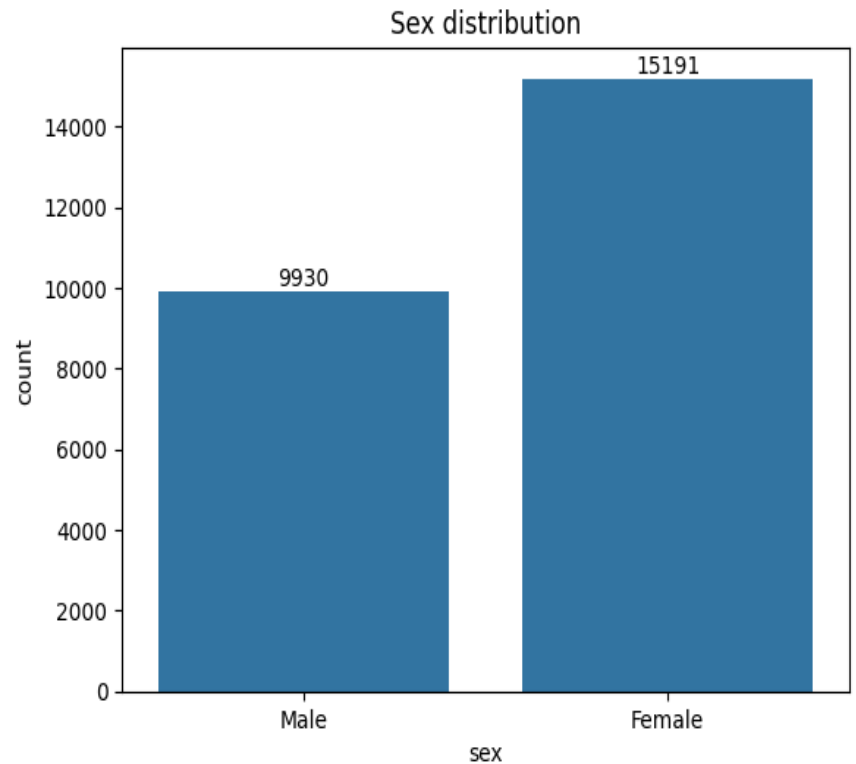
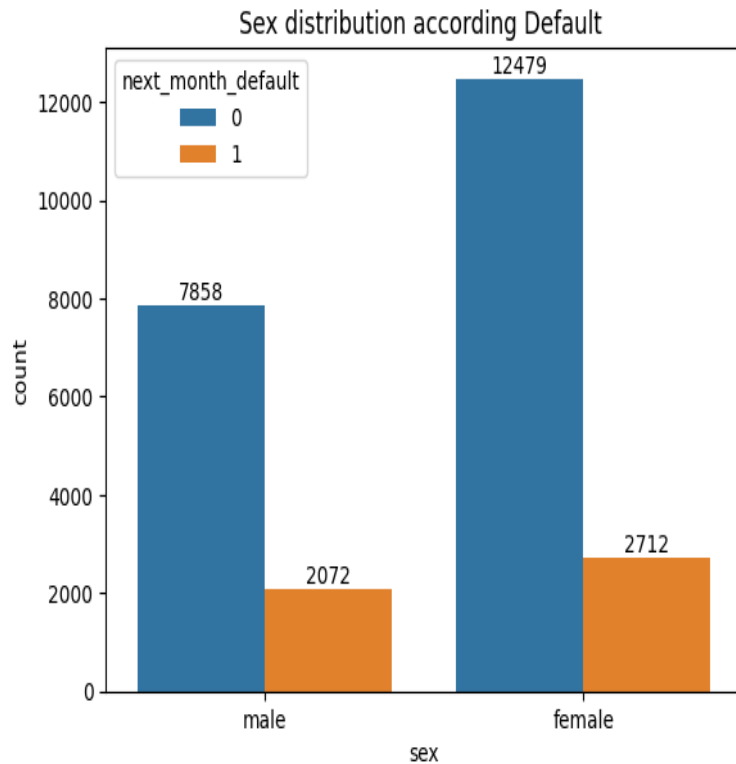
- **F2-Score prioritized** → catch as many defaulters as possible
- Trade-off → Slightly lower precision for **higher recall**

EDA Findings & Visualizations

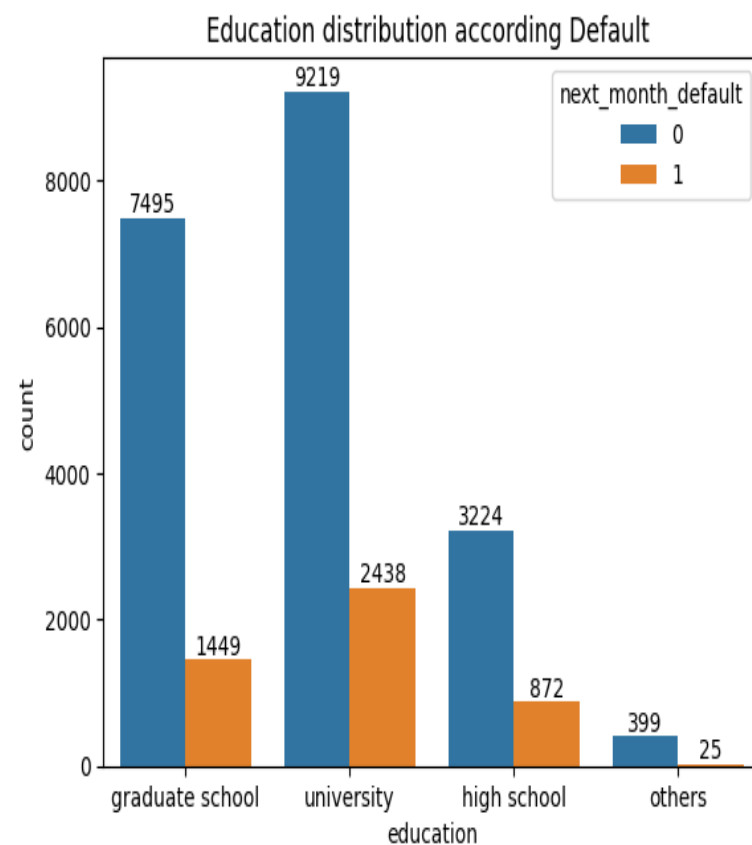
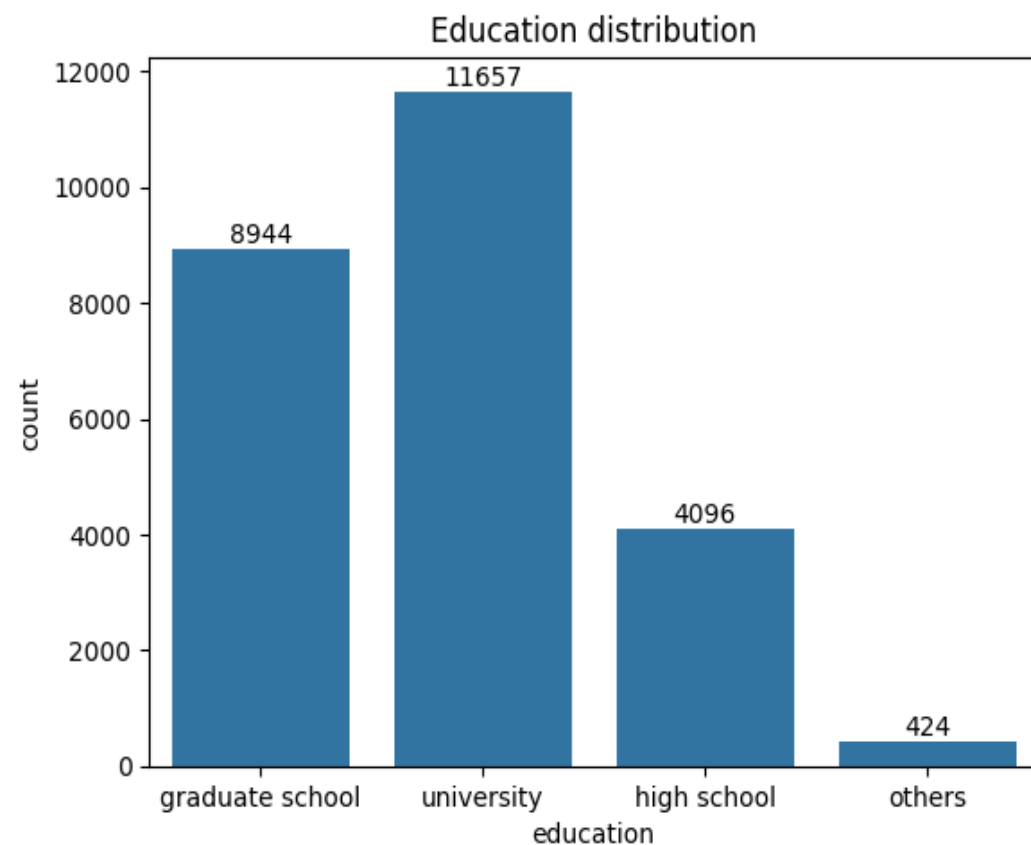
- Most defaults occur among customers with high bill amounts and past due payments.
- Correlations found between PAY_* variables and default risk.
- Class imbalance: ~19% defaulters, handled using SMOTE/SMOTE-Tomek.



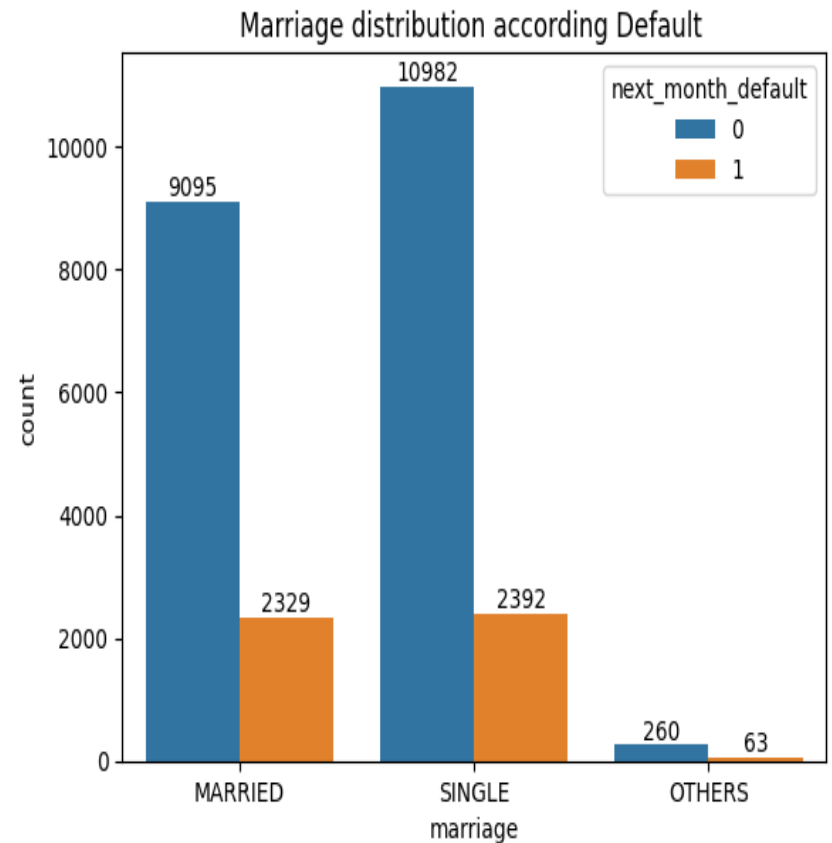
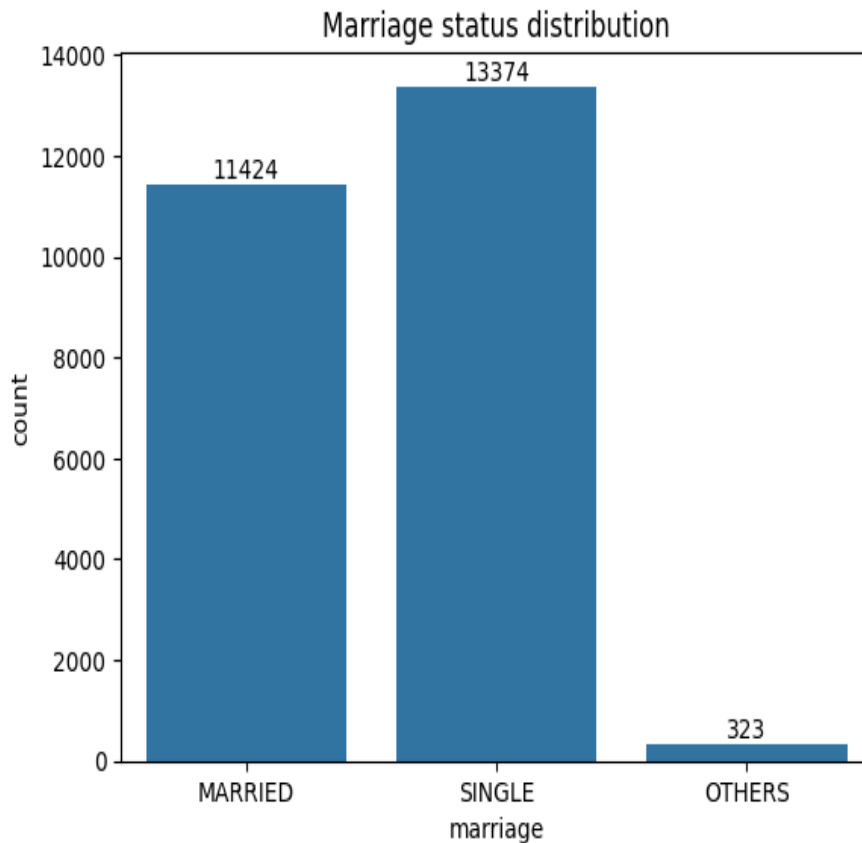
The data is quite imbalanced which about 19% of clients will default next month.



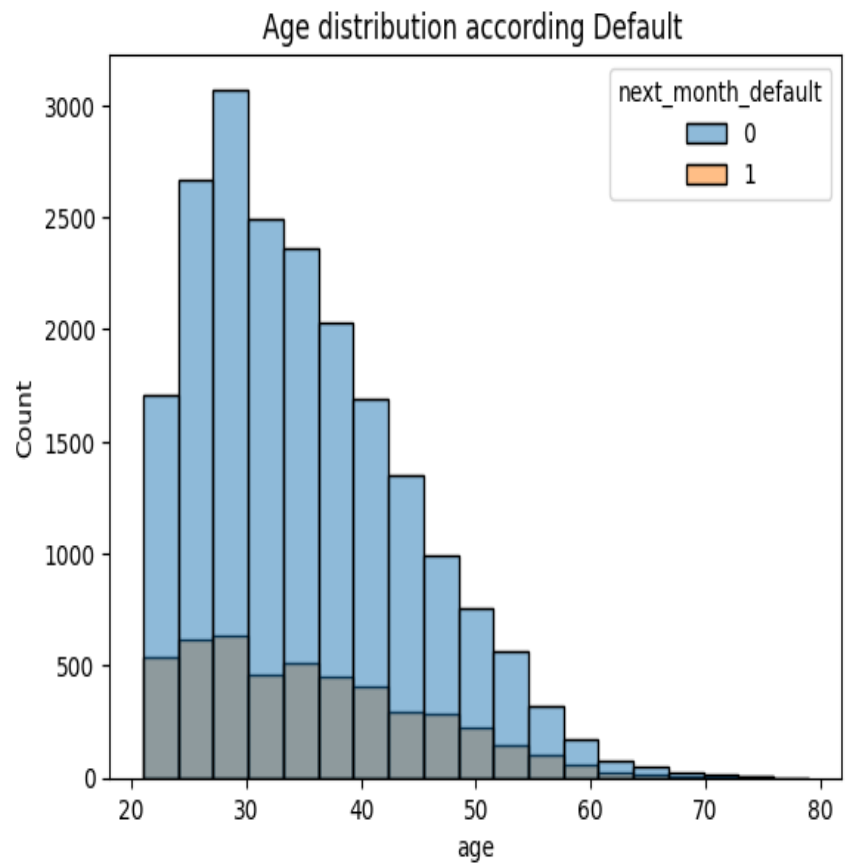
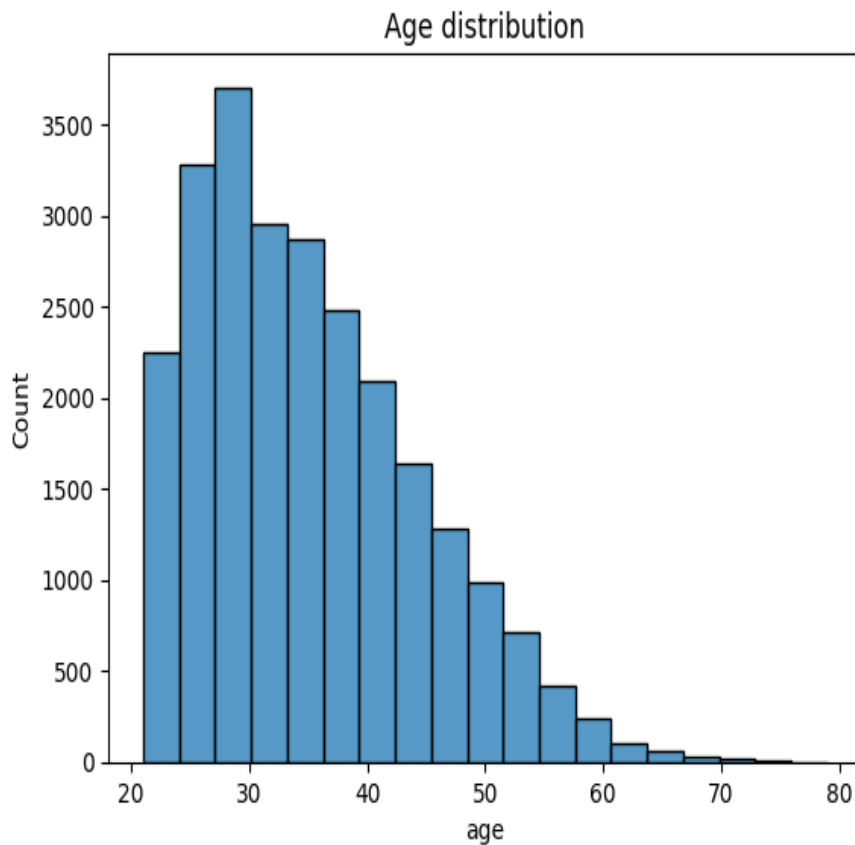
More Female clients own credit card than Male client. 21% of male clients fraud credit card while the ratio for female is around 18%.



University students are the group which highest number customers using credit cards (46%). High school students are the group which has highest fraud cases(21%), followed by university student(20%).

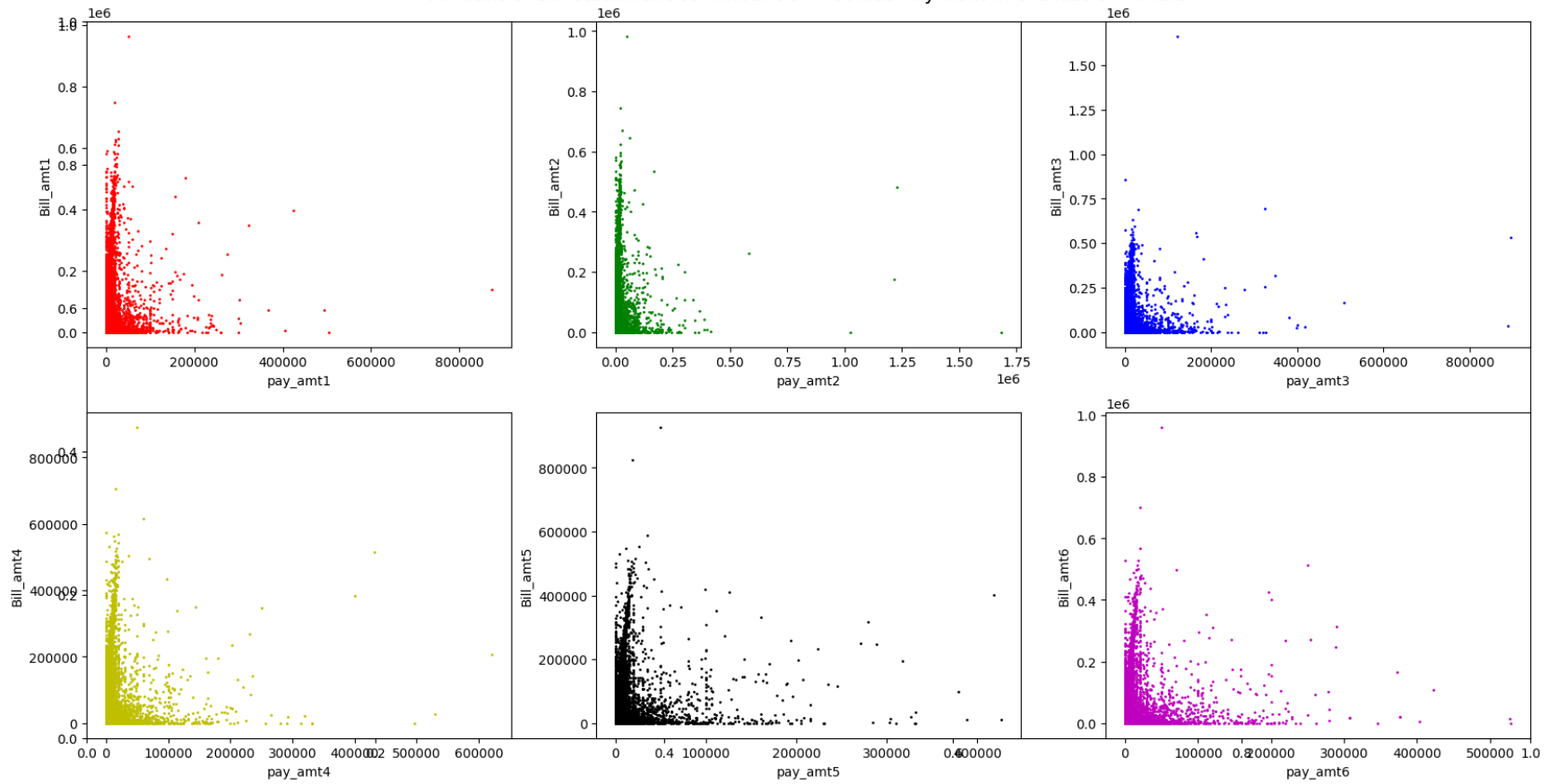


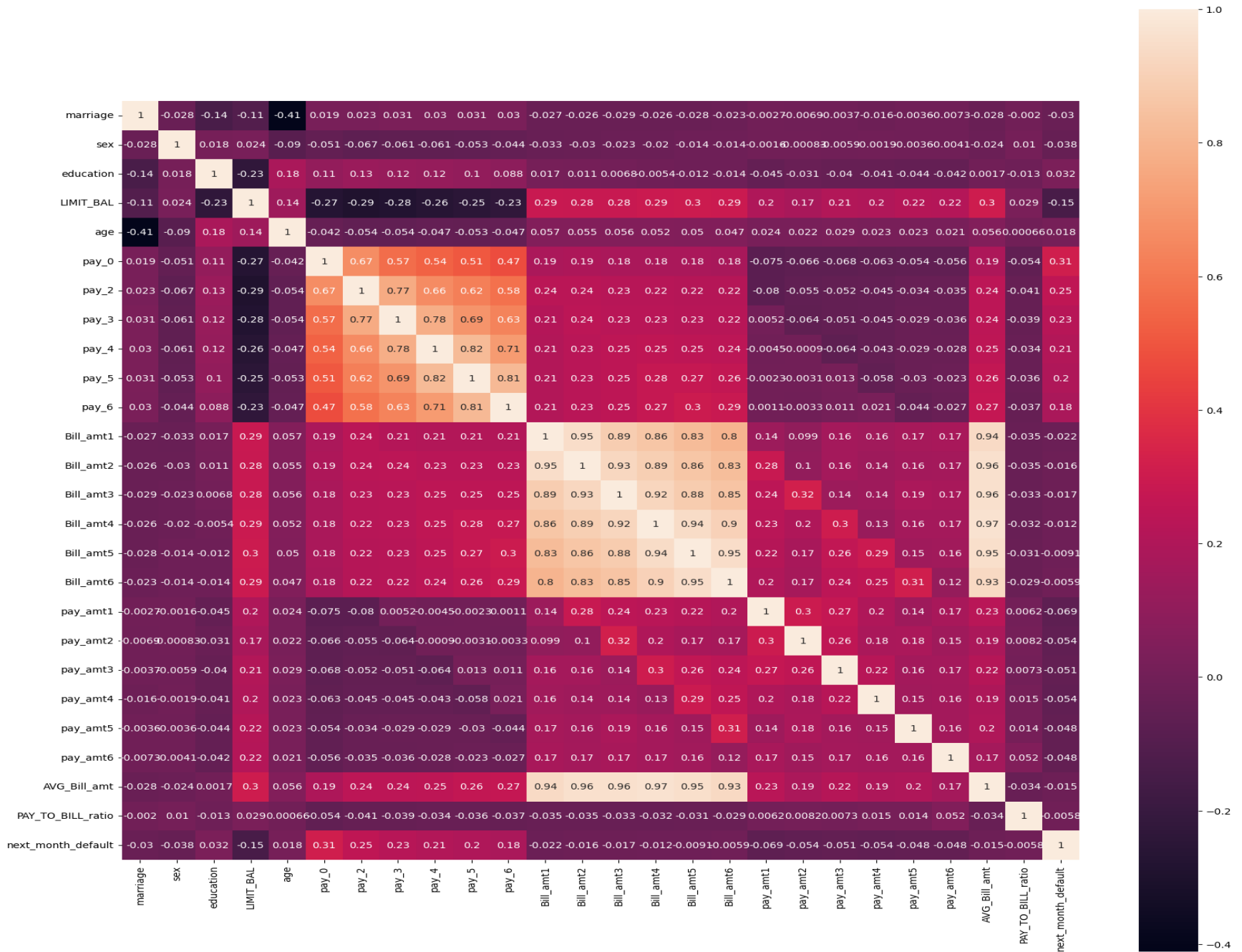
Single is the group which highest number of customers using credit cards (53%). Married people are the group which has highest fraud cases(20%). Others are not considered since they are very small in numbers.



The histogram is right-skewed, meaning that older customers are less likely to use credit cards.
The main client is in their 30s.
Customers in their 30s are also the most prone to credit fraud

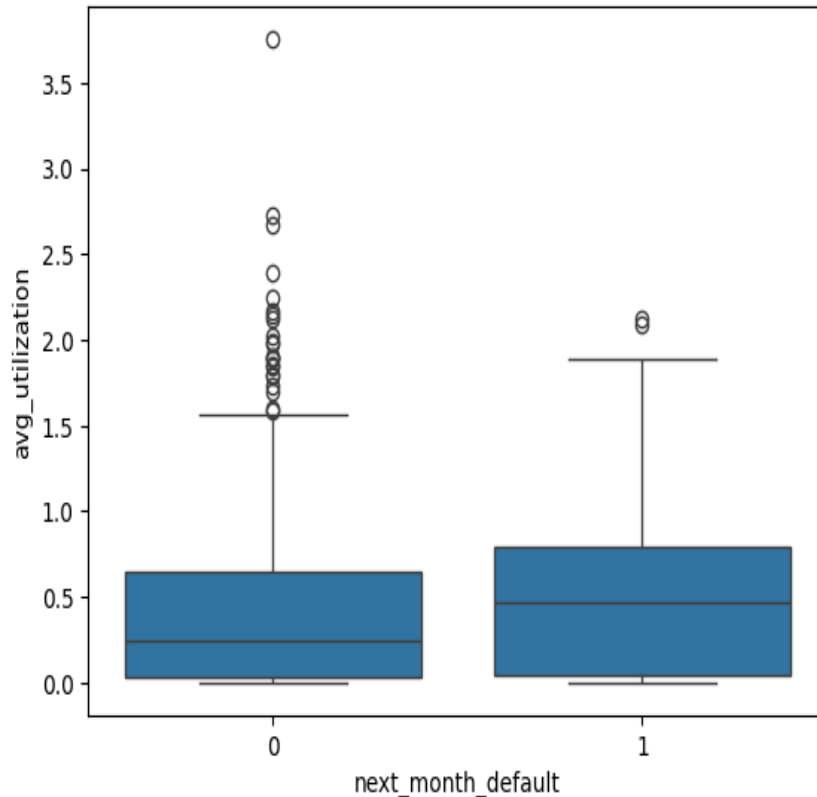
Amount of Bill statement vs Amount of Previous Payment in the last 6 months





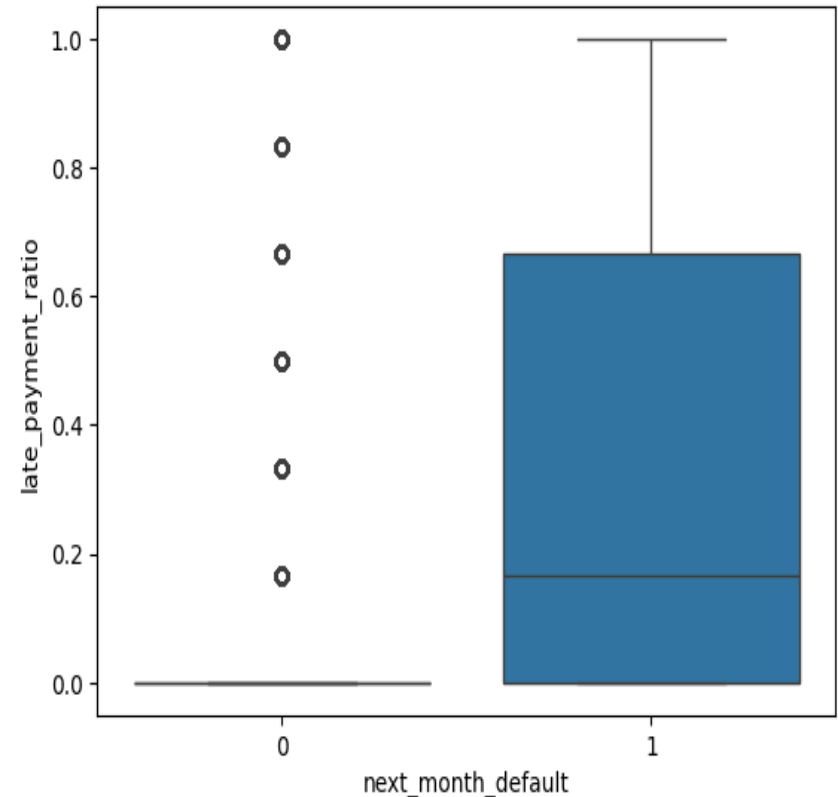
Correlation is high among PAY_0,2,3,4,5,6 and BILL_AMT1,2,3,4,5,6.

Utilization Rate vs Default

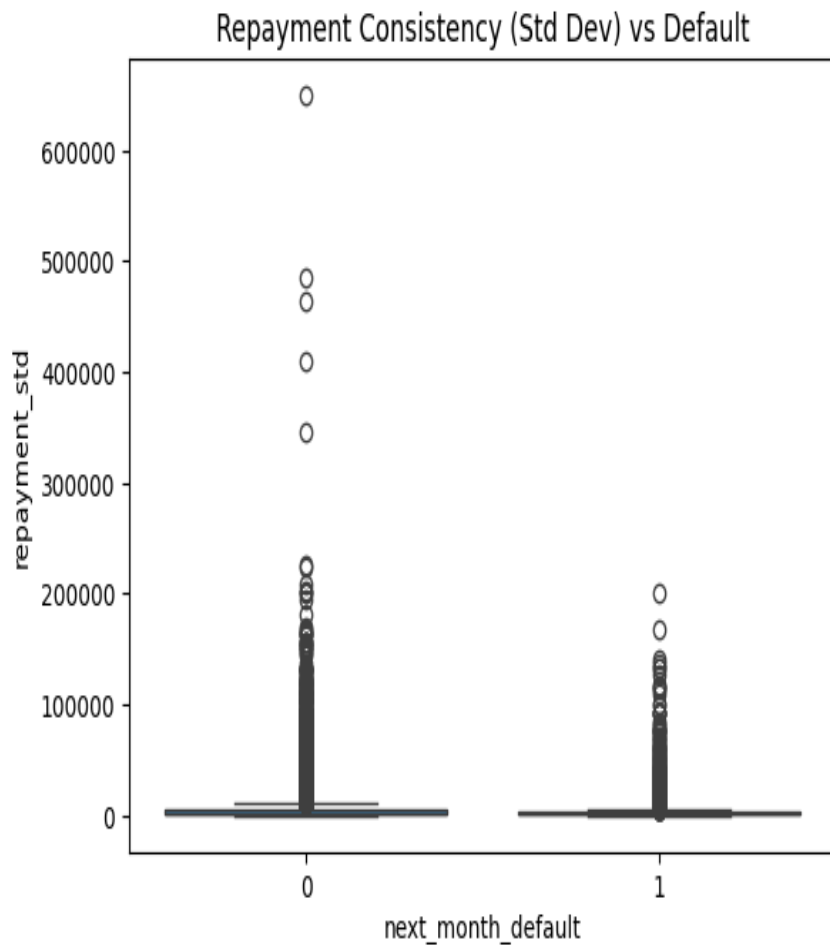


- Median utilization rate for non-defaulters (0) is around **0.2**, while for defaulters (1) it rises to approximately **0.5**.
- Some non-defaulters have utilization rates >3.5, showing extreme credit usage.

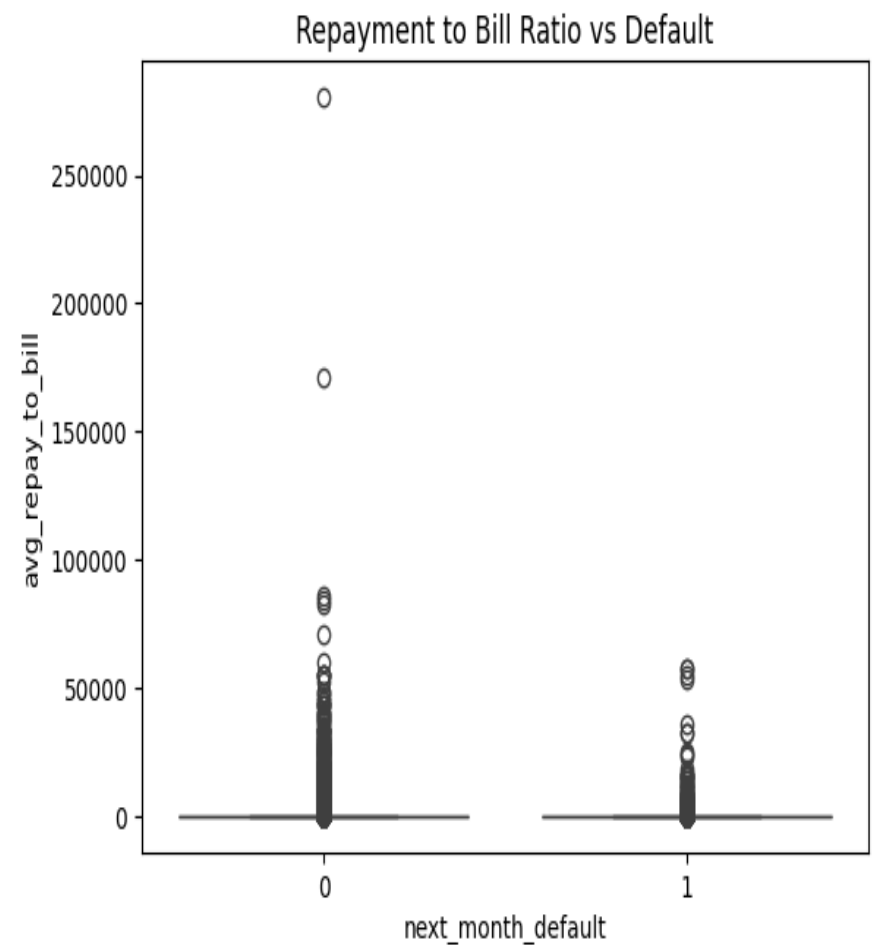
Late Payment Ratio vs Default



- Median late payment ratio for non-defaulters is **0.0**, while defaulters have a median of around **0.17**.
- Defaulters frequently have late payment ratios between **0.5 to 1.0**, indicating habitual late payment behavior.



- Median repayment standard deviation for both groups is **near 0**, indicating most customers repay consistently.
- Some repayment std devs reach **600,000+** among non-defaulters, likely driven by very large repayment amounts.



- Median repayment-to-bill ratio is close to 0 for both groups, with some extreme outliers exceeding 250,000 for non-defaulters.

Financial Insights

1. Overdue Payments → Primary Driver

- **Higher number of delayed payments** → Strongly linked to default
- **Longest Overdue Streak** → Customers with repeated consecutive delays = **high risk**
- **Recent overdue history** → Even more predictive than older late payments

2. Credit Utilization Behavior

- **High utilization ratio** → Spending close to or over the credit limit → Indicator of **financial stress**

3. Repayment Patterns

- **Repayment Variability (repayment_std)** → Inconsistent repayments → Early signal of financial instability
- **Low repayment-to-bill ratios** → Customers paying small fractions of their outstanding bills → Higher default likelihood

4. Interaction Effects

- Features like **Utilization × Payment Ratios** → Helped capture risky profiles combining poor repayment with overuse of credit

Model Selection

- **Models Compared:**
- Random Forest
- LightGBM
- XGBoost
- **Performance Overview:**

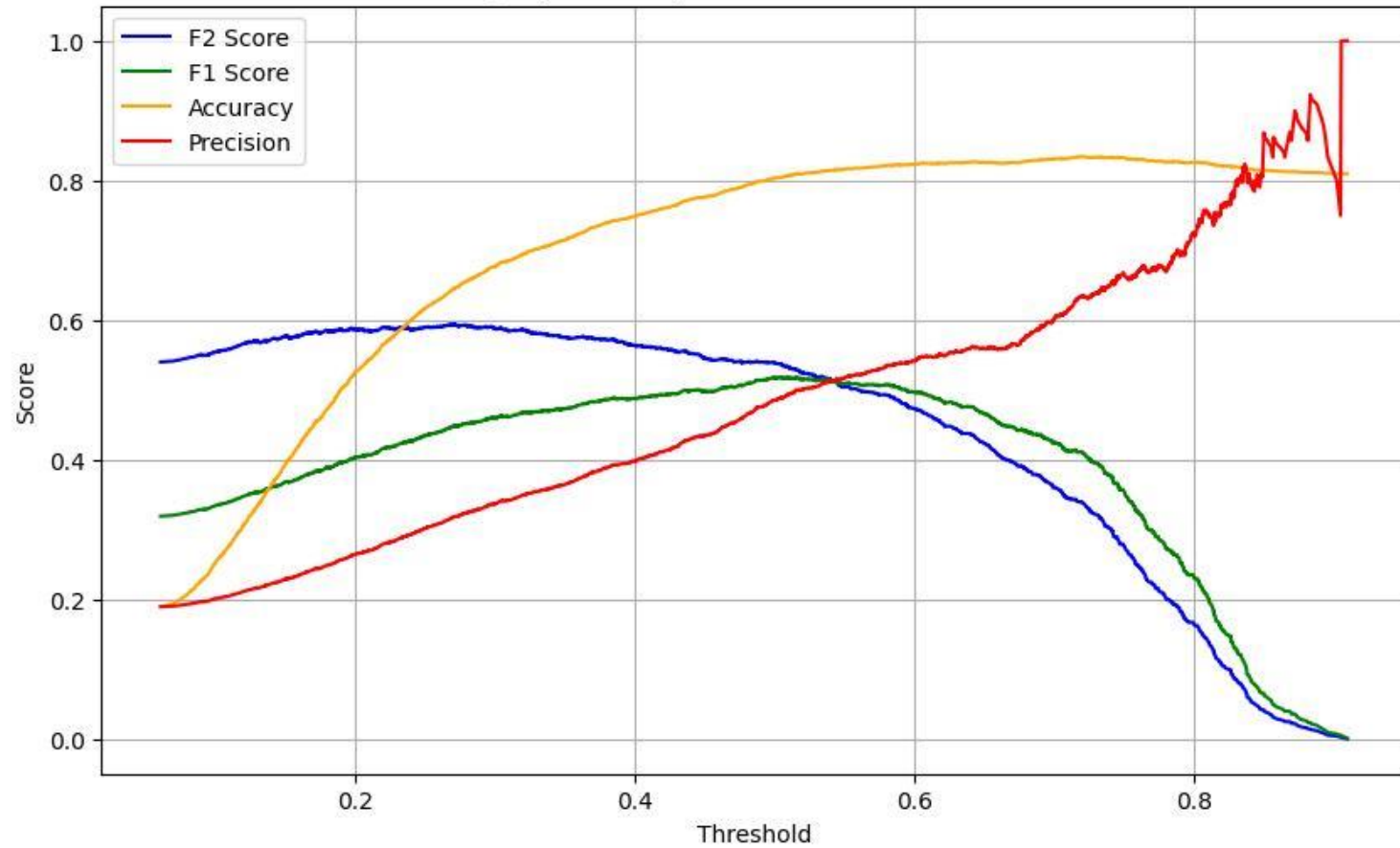
Model	Accuracy	AUC	F2 Score
Random Forest	0.804	0.767	0.540
XGBoost	0.821	0.763	0.431
LightGBM	0.828	0.767	0.423

- **Final Choice → Random Forest**
- **Highest F2-score (0.54) → aligned with credit risk priority (minimizing missed defaulters)**
- **Good AUC (0.77) → strong class separation ability**
- **⚖️ Acceptable trade-off in accuracy (80%) for improved recall of defaulters**
- Other models (XGB, LGBM) provided slightly higher accuracy but **lower F2**, meaning **more missed defaulters**, which is undesirable for credit risk.
- **Final Decision → Random Forest chosen for superior balance of recall and practical credit risk alignment.**

Evaluation Methodology

- **Accuracy** not suitable → Data is **imbalanced**.
- Prioritized **F2-score** → Focus on **catching defaulters**.
- Why F2? → **Minimizes false negatives**, aligns with credit risk priorities.
- **AUC & Recall** used for support.
- → Goal: Reduce risk of missed defaulters, **even if some false positives occur**.

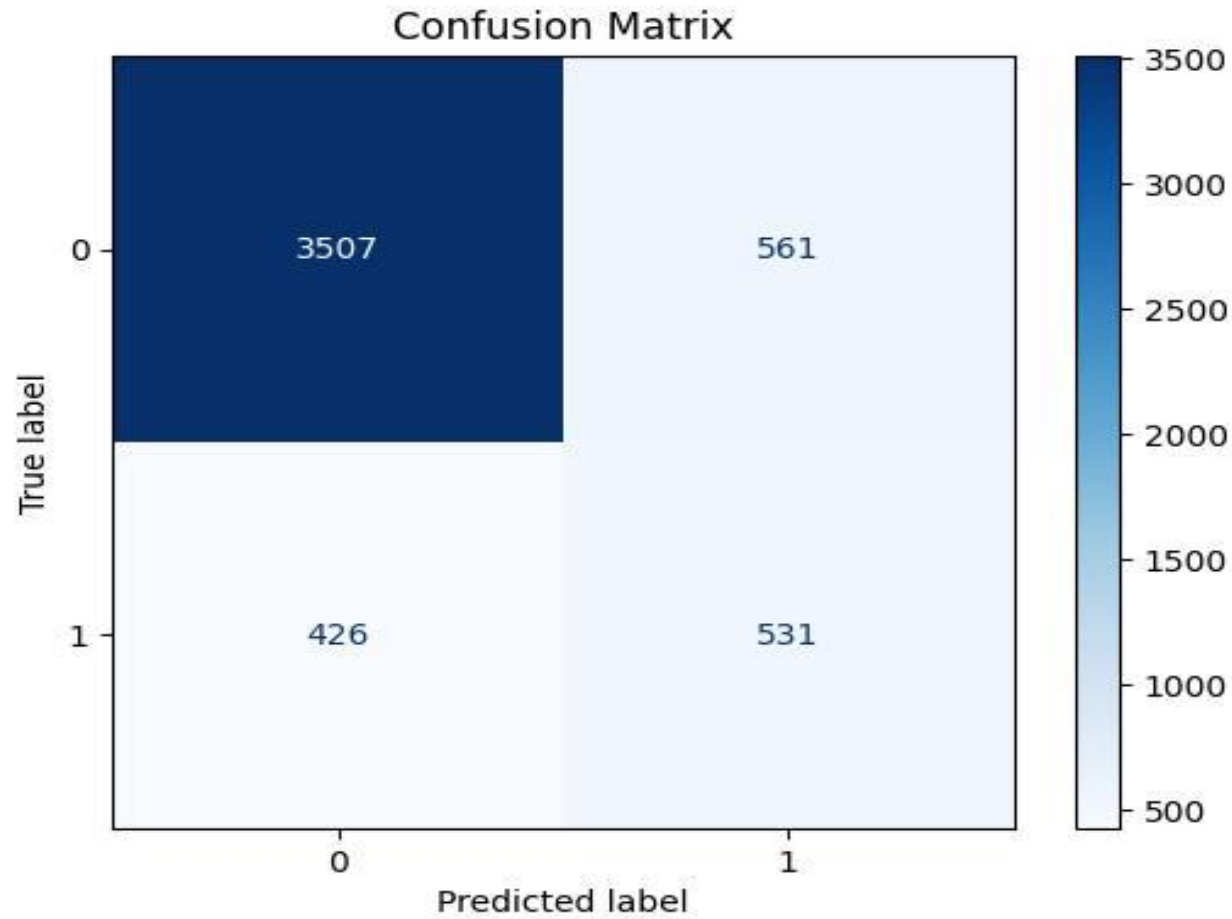
F2, F1, Accuracy & Precision vs. Threshold



Evaluation Metrics (Train Set)

- Model: Random Forest
- Accuracy: ~80.36%
- F1 Score: ~0.52
- Recall: ~0.70 (High recall ensures we capture most defaulters)
- F2 Score: ~0.54 (Recall-focused metric prioritized)
- AUC: ~0.77

Confusion Matrix



Classification Cutoff Selection

- After threshold optimization trials, we selected the **default cutoff of 0.5** for the final model.
- → **Why?** → **Provided a balanced trade-off** between accuracy (80.4%) and F2-score (0.54), without unnecessarily increasing false positives.
- **Custom threshold tuning provided only marginal F2 improvements** while significantly hurting accuracy.

Conclusion:

- → **Threshold = 0.5** retained for **consistent, interpretable, and operationally practical results.**

Business Implications

- **Improved risk scoring** leads to **better credit** approvals.
- **Lower false negatives reduce losses** from **bad debt**.
- Allows for proactive risk-based customer segmentation.
- Operational Impacts of False Positives (Some **good customers** may be **flagged**; **borderline** cases should undergo **manual review** by credit teams).

Summary of Key Findings

- Overdue payments and high utilization are key risk drivers.
- Class imbalance impacts performance; oversampling helps.
- LightGBM gave optimal results with high recall and good precision.
- Understanding customer repayment behavior is critical to credit risk.

Key Learnings

- Repayment Behaviour is the Strongest Predictor.
- Class Imbalance Requires Special Handling.
- Overfitting is a Risk with Complex Models.
- Trade-Off Between Precision and Recall Must Be Managed.
- Threshold Optimization Improves Performance.

THANK YOU